



In Silico Identification of MYB Transcription Factors Involved in Mangosteen Anthocyanin and Xanthone Production

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Abstract

Mangosteen is valued for its health-promoting properties, primarily due to its rind, which is rich in phenolic compounds such as xanthenes and anthocyanins. These compounds exhibit notable pharmaceutical activities, including antioxidant, anti-inflammatory, and antimicrobial effects. Their concentrations peak during fruit ripening, a process in which R2R3-MYB transcription factors are known to play key regulatory roles, particularly in anthocyanin biosynthesis. To date, only one MYB gene, *GmMYB10*, has been experimentally validated as a regulator of anthocyanin production in mangosteen, while the regulatory mechanism underlying xanthone biosynthesis remains unresolved. Given that fruit ripening is a complex molecular process involving the coordinated expression of numerous genes and transcription factors, identifying additional MYB regulators is essential for enhancing phenolic compound production through biotechnological approaches. The availability of a reference transcriptome and prior gene expression datasets has enabled the application of weighted gene co-expression network analysis (WGCNA) to explore these regulatory networks. This study represents the first WGCNA performed on mangosteen, aiming to identify MYB transcription factors associated with phenolic compound biosynthesis during fruit ripening. From 46,681 genes analyzed, 50 co-expression modules were generated. Among these, three modules—plum2, darkturquoise, and pink—were significantly associated with fruit ripening stages 2 and 6. All three modules were enriched with genes related to xanthone biosynthesis, while the darkturquoise module also showed enrichment for phenylpropanoid biosynthesis. A total of 91 R2R3-MYB genes were identified, with seven localized within the darkturquoise and pink modules. Notably, DN34818, a hub gene co-expressed with *GmMYB10* and other phenolic biosynthesis-related genes, is predicted to function as an anthocyanin repressor. Conversely, DN13742, a hub gene in the pink module, is putatively involved as an activator of primary phenylpropanoid metabolism. These findings provide new insights into the regulatory networks controlling bioactive metabolite production in a tropical fruit and lay the foundation for functional validation and molecular breeding efforts aimed at improving the pharmaceutical value of mangosteen.

Keywords Anthocyanins · Fruit ripening · Mangosteen · R2R3-MYBs · WGCNA · Xanthenes

Introduction

Mangosteen (*Garcinia mangostana* L.) is renowned for its rich content of phenolic compounds, particularly anthocyanins and xanthenes, which are predominantly concentrated in the rind (pericarp). These metabolites exhibit potent antibacterial, antiviral, and antioxidant properties (Aizat et al. 2019; El-Seedi et al. 2010; Ovalle-Magallanes et al. 2017; Rizaldy et al. 2021; Shan et al. 2011), contributing considerable pharmaceutical and economic value to mangosteen. Xanthenes are often added to healthcare products

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and functional food, including fortified juices. Commercial mangosteen juice products such as XanGo (XanGo LLC, Orem, UT) and Mangoxan (Pure Fruit Technologies, American Fork, UT) are supplemented with essential minerals and marketed as dietary supplements (Marcason 2006). Despite its growing global demand, with an estimated 0.23 million tonnes of mangosteen exports reported in the Preliminary Results 2023 of the Major Tropical Fruits Market Review (FAO 2024), the molecular mechanisms underlying the phenolics biosynthesis in mangosteen remain poorly elucidated.

Both anthocyanins (Zhao and Tao 2015) and xanthones (Badiali et al. 2023) are derived from the phenylalanine-dependent pathway. While anthocyanin biosynthesis is well-characterized in model plants and crops (Liu et al. 2018a, b; Wu et al. 2023) and several biosynthetic genes have been validated in mangosteen (Palapol et al. 2009a, b), the regulatory network controlling the biosynthesis of phenolic compounds, especially at the transcription level, is still underexplored. In plants, these pathways are commonly regulated by MYB transcription factors (TFs), particularly the R2R3-MYB subfamily (Pratyusha and Sarada 2022), which serves as central regulators of anthocyanin production (Yang et al. 2022).

Mangosteen undergoes seven fruit ripening stages (Osman and Milan 2006) (Fig. S1), with anthocyanin and xanthone contents reaching their peak at the final stage (Khaw et al. 2020; Palapol et al. 2009a, b). At stage 0, the rind is greenish-yellow with red spots, progressing to a pale yellowish-pink with pink speckles at stage 2. By stage 3, the rind turns uniformly pink, marking to optimal stage for export. Stage 4 is characterized by a red to brownish-red rind, while stage 5 displays a red-purple hue, indicating the fruit is suitable for consumption. At the final stage 6, the rind becomes purple, deep purple, or nearly black, coinciding with the highest accumulation of anthocyanins (Palapol et al. 2009a, b). Unlike typical climacteric fruits, mangosteen will only continue to ripen if harvested from at least stage 2 onwards (Abdul-Rahman et al. 2017).

To investigate molecular changes and regulatory pathways during mangosteen fruit ripening, a previous study conducted sequential pairwise differential expression analysis across ripening stages (S0 vs. S2 and S2 vs. S6) (Jamil et al. 2023). Recently, a mangosteen reference transcriptome was generated and became accessible through the online mangosteen eFP browser (https://bar.utoronto.ca/efp_mangosteen/cgi-bin/efpWeb.cgi), allowing researchers to visualize gene expression profiles across different tissues and developmental stages to support functional genomics research (Wee et al. 2024).

However, transcriptomic profiling and differential expression analysis alone provide a limited, gene-centric perspective and are insufficient for capturing the complex

regulatory interactions underlying fruit ripening (Horvath and Langfelder 2011). Hence, network-based approaches, particularly gene co-expression network analysis, have gained popularity for elucidating transcriptional regulation during fruit development (Wee et al. 2023). The network analysis facilitates the identification of key regulators associated with specific metabolic pathways through hub genes (Langfelder and Horvath 2008). For instance, hub genes involved in ester accumulation in strawberries (Li et al. 2020), carotenoid metabolism in apricots (Zhang et al. 2019a, b), fatty acid metabolism in pears (Li et al. 2022a, b), and high-quality fiber production in upland cotton (Zou et al. 2019).

Genetic engineering of mangosteen fruit is currently unfeasible, and fruits are typically harvested from mature trees with variable chromosome numbers (from 74 to 110) and complex genomic structures (Midin et al. 2018). Therefore, related genes must be identified through transcriptome profiling and weighted gene co-expression network analysis (WGCNA) to infer genes associated with key metabolic pathways. In this study on mangosteen postharvest biology, all publicly available RNA-seq datasets were used for WGCNA to identify MYB TFs involved in the biosynthesis of phenolic compounds. Given that ripening influences anthocyanin and xanthone accumulation, co-expression modules associated with different ripening stages were analyzed alongside KEGG pathway enrichment for phenolic biosynthesis. This approach enabled the identification of candidate regulator genes informed by both co-expression patterns and differential expression analysis.

Materials and Methods

Data Resources and Reference Transcriptome

This study was based on publicly available RNA sequencing data of mangosteen. A total of 56 short-read sequencing datasets were previously generated from different samples of mangosteen (Table 1 and Table S1) by Universiti Kebangsaan Malaysia (UKM), Malaysia and Bogor Agricultural University, Indonesia. Illumina sequencing technology was used to sequence the mangosteen samples of variety Manggis from UKM, compared to that of Ion Torrent sequencing performed by researchers from Bogor Agricultural University. Details on the samples used and experimental treatments are shown in Table 1.

A *de novo* assembly of reference transcriptome was performed using Trinity v.2.9.1 (Grabherr et al. 2011) to assemble all the mangosteen Illumina datasets in public datasets (Wee et al. 2024). TIGR Gene Indices clustering tools (TGICL) (Perteau et al. 2003) was used to generate the

Table 1 Manggis RNA-seq data found in SRA/DRA databases

Sequencing platform	Experiment/Tissue	BioProject accession	No. of samples	Raw data (Gb)	References
Illumina	Seed development	PRJNA395504	8	68.0	Mazlan et al. 2018
	Fruit ripening	PRJNA339916	6	29.2	Jamil et al. 2023
	Seed germination	PRJNA321150	4	66.3	Goh et al. 2019
	Callus	PRJNA279326	2	43.5	Mahdavi-Darvari and Noor 2016
Ion Torrent*	Aril/rind	PRJDB5091	12	29.6	Matra et al. 2016
	Normal/abnormal aril/rind	PRJDB5091	24	51.4	Matra et al. 2019

*Data deposited in the DRA database

non-redundant unigenes (denoted as genes here onwards) as a reference transcriptome (Wee et al. 2024). As there were two sequencing strategies, the normalization of expression profiling using the reference transcriptome was based on transcripts per million (TPM) values for subsequent network analysis. Details on the read statistics, data pre-processing, gene expression estimation, and the reference transcriptome have been described previously (Wee et al. 2024).

Principal Component Analysis and Differentially Expressed Gene Analysis

The principal component analysis (PCA) plot of all RNA-seq datasets was generated using the script 'PtR' (default parameter) found in the Trinity toolkit. The samples were grouped into four clusters corresponding to different tissues (Fig. S2). Illumina sequence datasets from germinating seed samples clustered more closely with callus and fruit samples than with developing seeds, which in turn separated from the aril and rind samples from Ion Torrent sequencing. This could reflect the confounding effect of different sequencing platforms or represent the differences between the different samples. Sample AR-4 was identified as an outlier in PCA (Fig. S2).

Two distinct ripening time points (S0 vs. S2 and S2 vs. S6) were used to perform differentially expressed gene (DEG) analysis using DESeq2 (Love et al. 2014) based on Salmon (Patro et al. 2017), the alignment-free abundance estimation method. DEGs were identified when the false discovery rate (FDR) was <0.001 and the $\log_2$ fold change ($\log_2FC$) values >2 or <-2 . The gene ontology (GO) classification of DEGs was visualized using the WEGO 2.0 online tool (Ye et al. 2018). The enriched KEGG pathways of DEGs were determined based on enrichment factor (EF) >1.0 and $FDR <0.01$.

Weighted Gene Co-Expression Network (WGCNA) Analysis

Data Processing and Network Analysis

Following PCA, data filtering was performed to select genes with average TPM >1 and coefficient of variation (CV) >1 . WGCNA v1.70 (Langfelder and Horvath 2008) was used for network analysis. As part of the preprocessing for network construction, hierarchical clustering of samples was conducted. Outlier detection was performed using a distance threshold of 200,000; samples exceeding this threshold were considered outliers. Sample AR-4 exceeded this threshold (Fig. S3) and was therefore excluded from downstream analysis. A soft-thresholding power of 6 was selected based on scale-free topology criteria (Fig. S4). The network was then constructed using the blockwiseModules function with signed topological overlap, a minimum module size of 30, and a module merging threshold (mergeCutHeight) of 0.25 (Chen et al. 2021; Langfelder and Horvath 2008; Liu et al. 2021). All other parameters were set to default values.

Selection of Target Modules Related to Fruit-Ripening and Screening for Hub Genes

Module-trait (tissue type) associations were quantified using the eigengene profile for each module to identify fruit ripening modules (S2 and S6). Next, the selected modules were used to perform KEGG pathway enrichment analysis based on the hypergeometric test in Microsoft Excel. To visualize the heatmap of enriched pathways (enrichment factor [EF] >1 and false discovery rate [FDR] <0.05 , adjusted using the Benjamini-Hochberg method), a free online tool, SRPlot (https://www.bioinformatics.com.cn/plot_basic_cluster_heatmap_plot_024_en), was used to draw the bubble plot. To identify xanthone-related genes, a local BLASTx

search was conducted using a manually curated list of xanthone-related proteins obtained from the published literature (Wee et al. 2024), available via figshare (<https://doi.org/10.6084/m9.figshare.23161250>). The software Perseus v1.6.15.0 was used to identify modules enriched with xanthone-related genes via Fisher's exact test ($EF > 1$ and $FDR < 0.05$, Benjamini-Hochberg adjusted).

In WGCNA, hub genes are typically defined as those with the highest connectivity within a co-expression module, often representing the top 10% of genes by connectivity (Liu et al. 2019). Differential expressions have also been used as an additional criterion for hub gene selection to ensure biological relevance (Zhang et al. 2023; Zong et al. 2023). In the present dataset, the connectivity values within modules showed only minimal variation among the most highly connected genes. To avoid the arbitrary exclusion of nearly equivalent genes, hub genes were identified as DEGs having at least 90% of the highest degree of connectivity within their respective module.

Interaction Network Generation

The gene interaction network was generated using WGCNA and exported to the Cytoscape 3.10.3 software (Shannon et al. 2003). Then, fruit ripening-related genes were identified. Discrete mapping was used to assign different fill colors for different fruit ripening traits. For the hub genes, the border color and the width were set as red and 10.0, respectively. The node label was set using passthrough mapping and a diamond shape was assigned for TF-related genes.

Identification of R2R3-MYB Transcription Factors

To identify the R2R3-MYB transcription factors within the co-expression modules of interest, a combination of domain- and motif-based approaches was employed. Functional annotation of the assembled transcriptome was performed using Trinotate pipeline (Bryant et al. 2017; Wee et al. 2024), available via figshare (<https://doi.org/10.6084/m9.figshare.23161250>), incorporating outputs from HMMER v3.0, which was used to identify protein domains by querying the PFAM (Release 2020–05–02). MYB domain-containing proteins were identified based on the presence of the full-length MYB DNA Binding Domain (DBD) (PFAM accession PF00249), using an E-value threshold of $< 1e-05$ (Khaksar and Sirikantaramas 2021; Wei et al. 2022). Next, these MYB TFs were subjected to motif analysis using the MEME v5.5.7 online tool (<http://meme-suite.org/index.html>) to identify conserved R2 and R3 motifs, which are characteristic of R2R3-type MYB transcription factors. The MEME analysis was performed using default parameters with minor adjustments, including zero or one occurrence

per sequence, motif width range of 6–60 amino acids, and a maximum number of 20 motifs. Only those MYB proteins containing both conserved R2 and R3 repeats were classified as R2R3-MYB TFs and subsequently identified from the co-expression modules of interest.

Phylogenetic Analysis

To understand the evolutionary relationship of R2R3-MYBs within the modules of interest in *G. mangostana* with other plants MYB TFs reported involved in flavonoid biosynthesis (Deng et al. 2018; Liu et al. 2015; Pandey et al. 2015), their protein sequences (Table S2) were downloaded from the National Center for Biotechnology Information (NCBI) database and aligned using the ClustalW tool in the BioEdit v.7.2.5 (Hall 1999) with default parameters. Then, a phylogenetic tree was generated using MEGA 11.0 (Tamura et al. 2021) with the neighbor-joining (NJ) method and 1,000 bootstrap replicates. There was a total of three mangosteen R2R3-MYB genes, *GmMYB1*, *GmMYB7*, and *GmMYB10* were reported in 2009 (Palapol et al. 2009a, b). The amino acid sequence of *GmMYB10* was 100% identical to DN65768, indicating they are the same gene. Therefore, both were grouped and named *GmMYB10/DN65768*.

Gene Expression Heatmap Generation

The expression heatmap of the fruit ripening-related genes and R2R3-MYB genes was generated using Perseus v1.6.15.0 software (Tyanova et al. 2016). The TPM values were Z-score normalized with the median values before generating the hierarchical clustering heatmap of the gene expression using Pearson correlation with default settings.

Results

Differentially Expressed Gene Analysis of Mangosteen Fruit Ripening

Mangosteen has a unique ripening behavior compared to other climacteric fruits that ripen even when harvested at the immature (S0) stage (Paul et al. 2012). Mangosteen can only be harvested at stage 2 (S2) of ripening to be fully ripened but not when the fruit is immature (S0) (Palapol et al. 2009a, b). Therefore, (Abdul-Rahman et al. 2017) performed RNA-seq for samples at stages 0, 2, and 6 of mangosteen fruit ripening. A transcriptomics analysis of mangosteen ripening across these stages was later conducted by (Jamil et al. 2023).

In the present study, differentially expressed genes (DEGs) were identified based on unigene expression data from the same three different ripening stages (S0, S2, and S6). In total, there were 3,731 DEGs (Table 2). The S0 vs. S2 comparison yielded the highest number of DEGs (3,122) of which 2,159 were downregulated. In the S2 vs. S6 comparison group, more genes (518) were upregulated among the 826 DEGs. The total number of downregulated genes (2,422) was higher than upregulated genes (1,469). Among the DEGs, 217 of them were shared among the comparison groups S0 vs. S2 and S2 vs. S6 (Fig. 1).

In comparison to the previous study (Jamil et al. 2023), which reported 6,233 differentially expressed transcripts (DETs), 535 DETs were found to be consistently expressed across both early (S0 vs. S2) and late (S2 vs. S6) ripening stages. A total of 4,445 DETs were identified during early ripening, with 3,635 being downregulated. Similar to DEG analysis, most of the DETs (1,019) identified during late ripening were upregulated (Jamil et al. 2023).

For gene ontology (GO) term analysis, the DEGs were divided into three main categories: biological process (2,462), cellular component (2,519), and molecular function (2,492) (Fig. 2). Cellular process, metabolic process, and response to stimulus were the three major subcategories under the biological process. In comparison, DETs were predominantly associated with ‘cellular process’, ‘metabolic process’, and ‘biological regulation’ (Jamil et al. 2023). Under the cellular component, both cell part and cell had the highest number (2,312 each), whereas DETs

additionally included the term ‘organelle’ (Jamil et al. 2023). For molecular function, the main subcategories for DEGs were ‘binding’ (1,722) and ‘catalytic activity’ (1,544). In contrast, DETs also showed high representation of ‘hydrolase’ and ‘transporter’, in addition to ‘binding’ (Jamil et al. 2023). This showed that most DEGs were related to cells and involved in binding and catalytic activities of cellular process, metabolic process, and response to stimulus during fruit ripening.

KEGG pathway enrichment analysis of DEGs showed that the enriched pathways include: “biosynthesis of secondary metabolites”, “plant hormone signal transduction”, “metabolic pathways”, “starch and sucrose metabolism”, “microbial metabolism in diverse environments”, “photosynthesis”, “photosynthesis-antenna proteins” and “biosynthesis of various plant secondary metabolites” (Table S3). In comparison, pathway enrichment analysis of DETs using KEGG classified transcripts into four categories: ‘metabolism’, ‘environmental information processing’, ‘cellular processes’, and ‘organismal systems’ (Jamil et al. 2023).

The overall enrichment analysis results of DEGs were coherent with the previously reported analysis from the differentially expressed transcript (DET) analysis (Jamil et al. 2023), which was based on transcript-level expression using a transcriptome generated from fruit-only datasets. This partly explains the different numbers obtained between transcript-level DETs and the gene-level DEGs in the current study, which is based on a reference transcriptome that has incorporated more datasets (Wee et al. 2024). Nonetheless, the reproducible results give confidence for further interpretation of network analysis results based on these DEGs.

Table 2 Summary of DEG analysis

Comparison	All	Upregulated	Downregulated
S0 vs. S2	3,122	963	2,159
S2 vs. S6	826	518	308
All	3,731	1,469	2,422

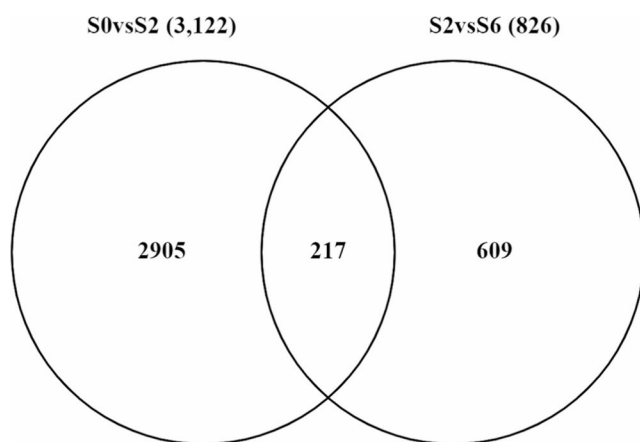


Fig. 1 Venn analysis of DEGs comparison during different fruit ripening time points. Parentheses indicate the total number of genes

Weighted Gene Co-Expression Network Analysis

A total of 46,681 genes (Supplementary File 1) with average TPM > 1 and coefficient of variation (CV > 1) were selected for the WGCNA tool (See Methods). Prior to network construction, the connection density was manually adjusted to determine the appropriate soft-thresholding power for the network. A low power makes the network dense with weak connections and as it becomes higher power, sparser. By choosing a soft-thresholding power of 6, WGCNA generated 50 densely connected modules (See Supplementary File 2 and Table S4 for the decision). Resulting network modules were automatically color-coded and we refer to modules by their colors. Among the 50 modules, the largest module was turquoise, comprising 8,870 genes, followed by the blue module with 7,735 genes. The smallest module, thistle2, contained 62 genes. 206 genes were not assigned to any module and were grouped into the grey module (Table S4).

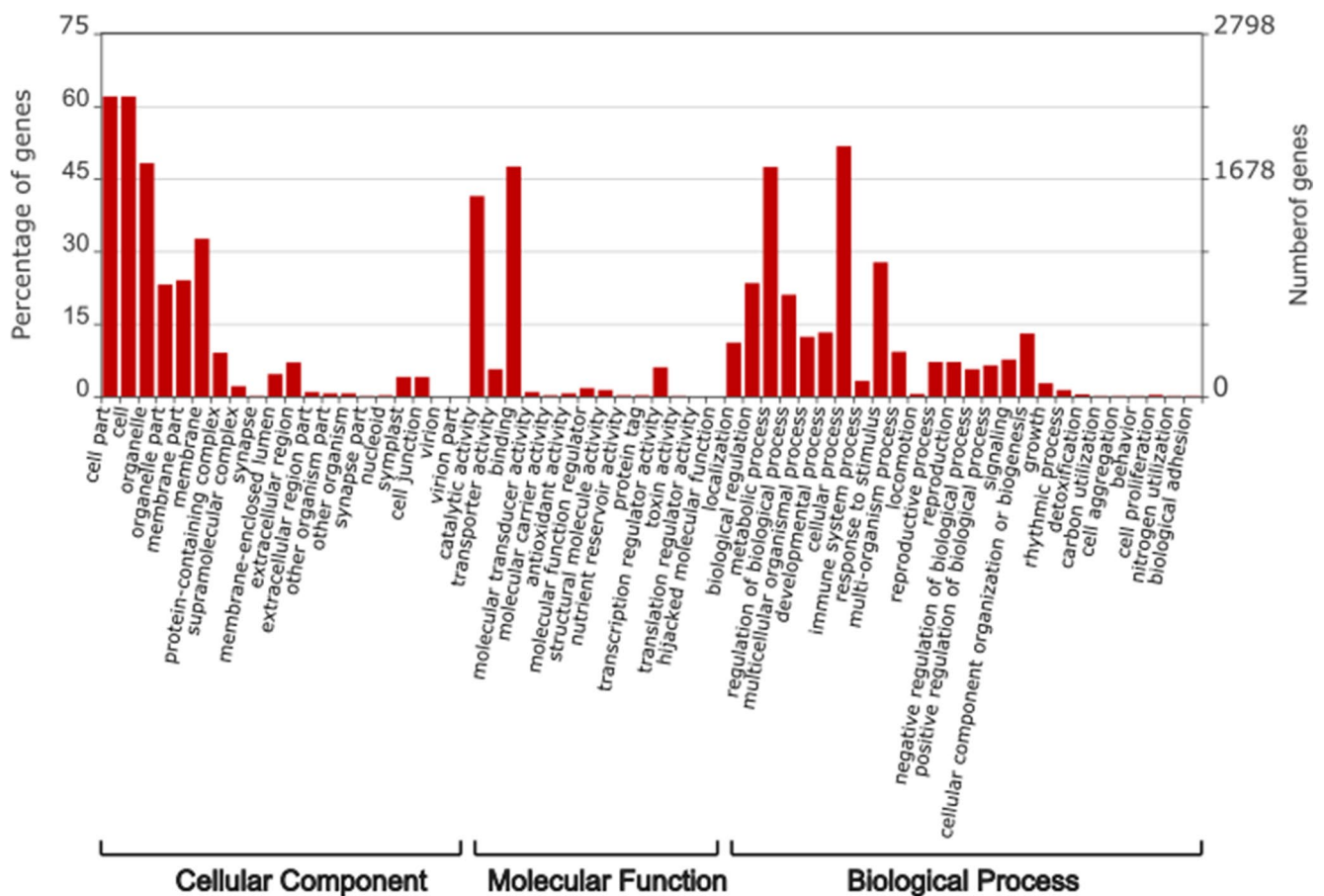


Fig. 2 Gene ontology (GO) term analysis of the DEGs

Screening of Fruit Ripening-Related Modules

To identify gene modules associated with fruit ripening, module eigengenes were correlated with developmental stages of the fruit. An eigengene is the first principal component of each cluster and represents the overall expression of the cluster. Three modules, pink (1,318 genes), plum2 (68 genes), and darkturquoise (503 genes), exhibited positive correlations with ripening stages S2 and S6 (Fig. S5), and were clustered accordingly (Fig. S6). Among these, the pink module showed the highest Pearson correlation coefficient (PCC) with the stage S2 (PCC=0.66), and the plum2 and pink modules showed strong correlations with stage S6 (PCC=0.64 and 0.56, respectively). These three modules were selected for further analysis. Functional annotation revealed that 74.6%, 86.8%, and 79.4% of the genes in the darkturquoise, plum2, and pink modules, respectively, were annotated (Supplementary Files 3–5). KEGG pathway enrichment analysis showed that only the darkturquoise module was significantly enriched with phenylpropanoid biosynthesis and other fruit ripening-related pathways, including “biosynthesis of secondary

metabolites”, “flavonoid biosynthesis”, “metabolic pathways”, “phenylpropanoid biosynthesis”, and “tropane, piperidine, and pyridine alkaloid biosynthesis” (Fig. 3).

Expression of Genes Involved in Xanthone and Anthocyanin Biosynthesis

Gene enrichment analysis revealed that all three modules were significantly enriched with xanthone-related genes. The plum2 module exhibited the highest enrichment factor (EF=20.03), followed by darkturquoise (EF=8.12), and pink (EF=5.55) (Table S5).

A summary of phenolic compound-related genes identified in the three modules is presented in Table 3. Most genes associated with anthocyanin biosynthesis showed higher expression during stage S2 in mangosteen ripening (Fig. 4A). Notably, all xanthone pathway-related genes detected in the plum2 module (Supplementary File 5) (except *DN47932-PAL*) were upregulated at S2 and peaked at S6, except for *DN4633-Ptpat* (Fig. 4B). Among these, *DN47932-PAL* and *BPS* were significantly upregulated during fruit ripening. Interestingly, *DN47932-PAL*

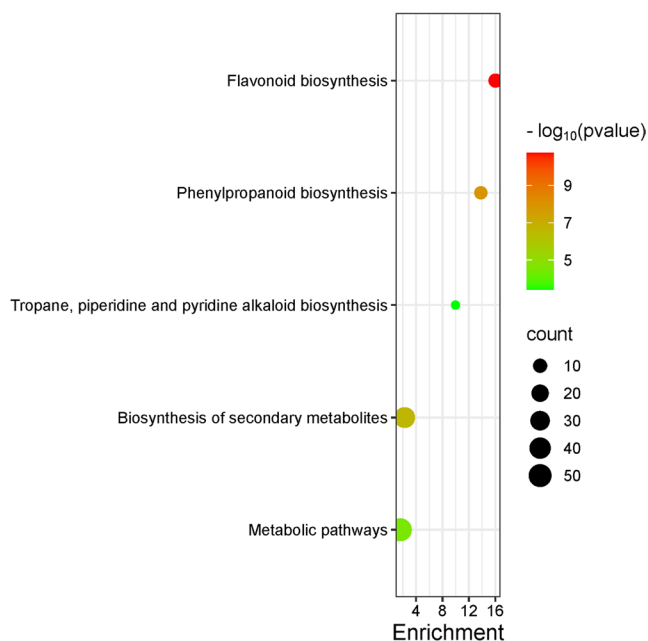


Fig. 3 Pathway enrichment in the darkturquoise module. Pathways shown have enrichment factor (EF) > 1 and FDR value < 0.05. The bubble sizes represent the number of genes while the colors indicate the $\log P$ -value

was downregulated in early fruit ripening (S0 vs. S2) and then significantly upregulated at S6 ($\log_2FC = 2.68$, $FDR = 3.09E-09$) (Supplementary File 5).

Identification of R2R3-MYB TFs

A total of ninety-one R2R3-MYBs with complete open reading frames (ORFs) ($e\text{-value} < 1e-05$) were identified from the mangosteen transcriptome data, with their corresponding amino acid sequences listed in Supplementary File 6. These sequences range in length from 123 amino acids (DN197641) to 1,030 amino acids (DN91653).

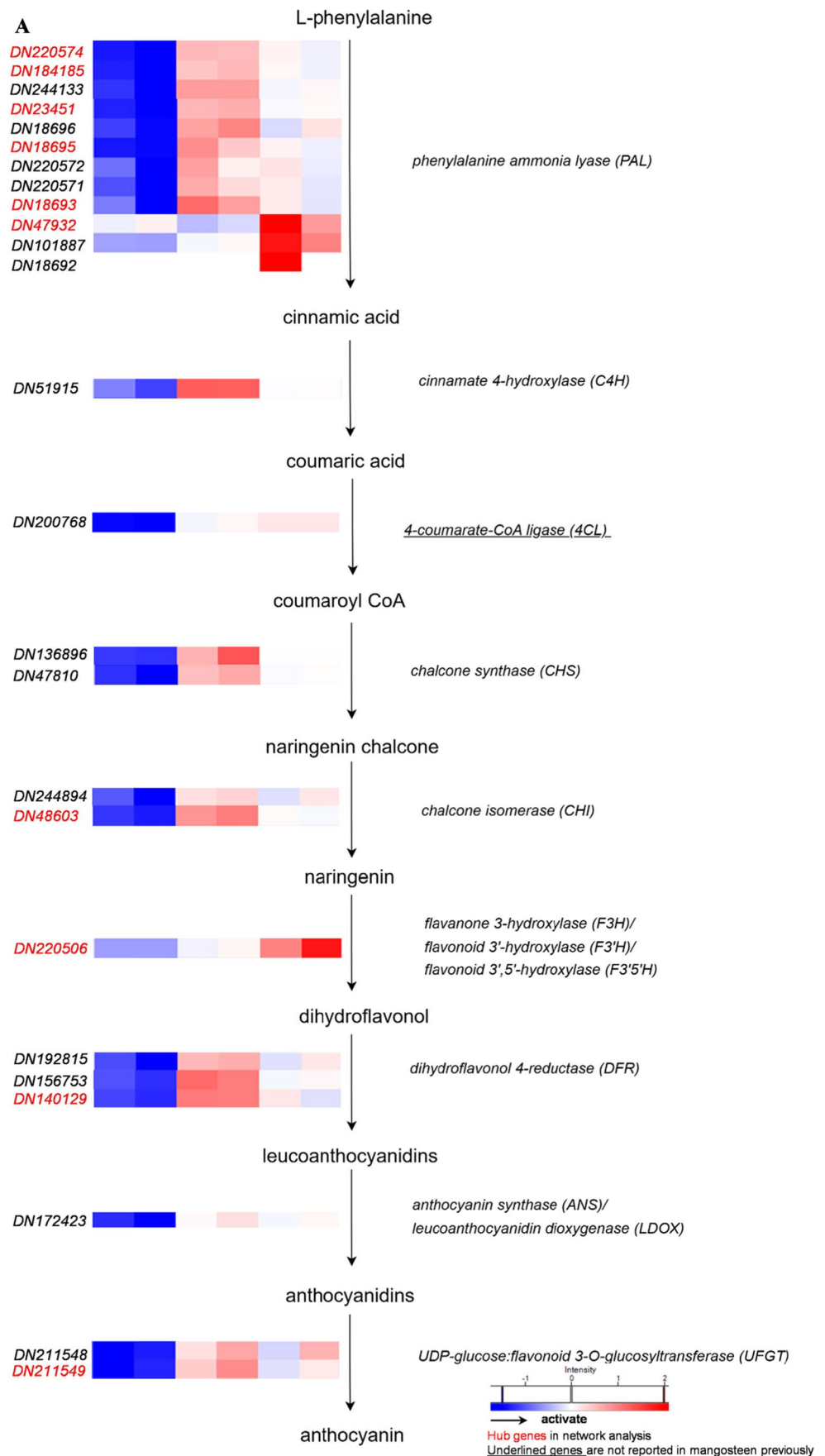
The data filtering step to select genes with average transcript per million (TPM) > 1 and coefficient of variation (CV) > 1 yielded a subset of 37 R2R3-MYB genes. Their expression profiles are shown in Fig. 5. Four R2R3-MYB TFs were identified within the darkturquoise module (DN65768, DN191051, DN34818, and DN37461), while 3 were found in the pink module (DN13742, DN182269, and DN47953). However, no R2R3-MYB TFs were detected in the plum2 module, despite its high enrichment with xanthone-related genes. Notably, DN37461 and DN47953 encode identical protein sequences but differ in their nucleotide sequences. Since the network analysis was based on nucleotide sequences, DN37461 was assigned to the darkturquoise module, while DN47953 was placed in the pink module. The majority of R2R3-MYB TFs in these two modules were upregulated at fruit ripening stage 2 (S2) (Fig.

Table 3 Phenolic compound-related genes identified in three modules of interest (darkturquoise, pink, and plum2)

Gene functions	Darkturquoise	Pink	Plum2
Phenylpropanoid pathway			
Phenylalanine ammonia lyase (PAL)	9	1	2
Trans-cinnamate 4-monooxygenase (CYP73A14)	1		
Cinnamate 4-hydroxylase (C4H)	1		
4-coumarate-CoA ligase (4CL)	1		
Flavonoid metabolism and biosynthesis			
Chalcone synthase (CHS)	1	1	
Chalcone isomerase (CHI)	2		
Flavonoid 3',5'-hydroxylase (F3'5'H)	1		
Flavonoid 3'-monooxygenase (CYP75B2)	1		
Naringenin,2-oxoglutarate 3-dioxygenase (AN3, F3H, FHT)	3		
Dihydroflavonol 4-reductase (DFR)	2	1	
Anthocyanin synthase (ANS)/leucoanthocyanin dioxygenase (LDOX)	1		
Anthocyanidin 3-O-glucosyltransferase (UGT)	1	1	
Flavonol synthase (FLS)	1		
Xanthone pathway			
Benzoate-CoA ligase (BZL)		1	
Benzophenone synthase (BPS)	1	4	1
1,3,5-trihydroxyxanthone synthase (1,3,5-triHXS)		1	
1,3,7-trihydroxyxanthone synthase (1,3,7-triHXS)	5	6	2
1,3,6,7-tetrahydroxyxanthone 8-prenyltransferase (Pt8px)		2	3
1,3,7-THX, 8-prenyl-1,3,6,7-tetrahydroxyxanthone 8-prenyltransferase (Ptpat)			2
Transporter genes			
Glutathione S-transferase (GST)	3	1	
Protein DETOXIFICATION 41 (DTX41)	1		
Fruit-ripening related transcription factors			
R2R3-Myeloblastis-related (MYB)	4	3	
Basic helix-loop-helix (bHLH)	1	2	
MCM1-AGAMOUS-DEFI-CIENS-SRF-box (MADS)	3	3	
WRKY transcription factors (WRKY)	1	3	
YABBY transcription factors (YAB)	2	2	
Ethylene-responsive transcription factor (ERF)			1
B3 domain-containing protein Os01g0234100			1

Numbers indicate gene counts in each module

Fig. 4 Anthocyanin and xanthone biosynthesis pathways. **(A)** Anthocyanin biosynthesis pathway. The phenylalanine-dependent pathway starts with the conversion of phenylalanine into cinnamic acid by phenylalanine ammonia lyase (PAL) (El-Mawla et al. 2001), followed by a series of enzymatic steps leading to anthocyanin formation. **(B)** Xanthone biosynthesis pathway. Phenylalanine is converted by PAL, followed by intermediate steps (not shown; indicated by a dotted arrow) leading to benzoic acid. Benzoic acid is then transformed into benzoyl-CoA by benzoyl-CoA ligase (BZL) (El-Mawla and Beerhues 2002). Benzoyl-CoA is converted to 2,4,6-trihydroxybenzophenone (2,4,6-triHB) by benzophenone synthase (BPS) (Nuallkaew et al. 2012) and then transformed into 2,3',4,6-tetrahydroxybenzophenone (2,3',4,6-tetraHB) by a cytochrome P450 (CYP) monooxygenase (El-Awaad et al. 2016) with benzophenone 3'-hydroxylase (B3'H) activity. The intermediate product, 2,3',4,6-tetraHB, undergoes regioselective oxidative C-O phenol couplings either *ortho* or *para* to the 3'-hydroxy group, ultimately yielding 1,3,5- and 1,3,7-trihydroxyxanthones (triHXs), respectively. These reactions are catalyzed by the enzymes 1,3,5-trihydroxyxanthone synthase (1,3,5-THX) and 1,3,7-trihydroxyxanthone synthase (1,3,7-THX synthase) (Khatab and Farag 2020). Prenyltransferase then catalyzes the conversion of 1,3,7-triHXs into α -mangostin (Khatab and Farag 2020). Heatmap displaying Z-score normalized TPM values. Red represents high expression, while blue represents low expression. Each column represents a sample replicate: S0_1; S0_2; S2_1; S2_2; S6_1; S6_2, respectively. Genes in red font are hub genes identified from network analysis while underlined genes are newly identified genes in this study



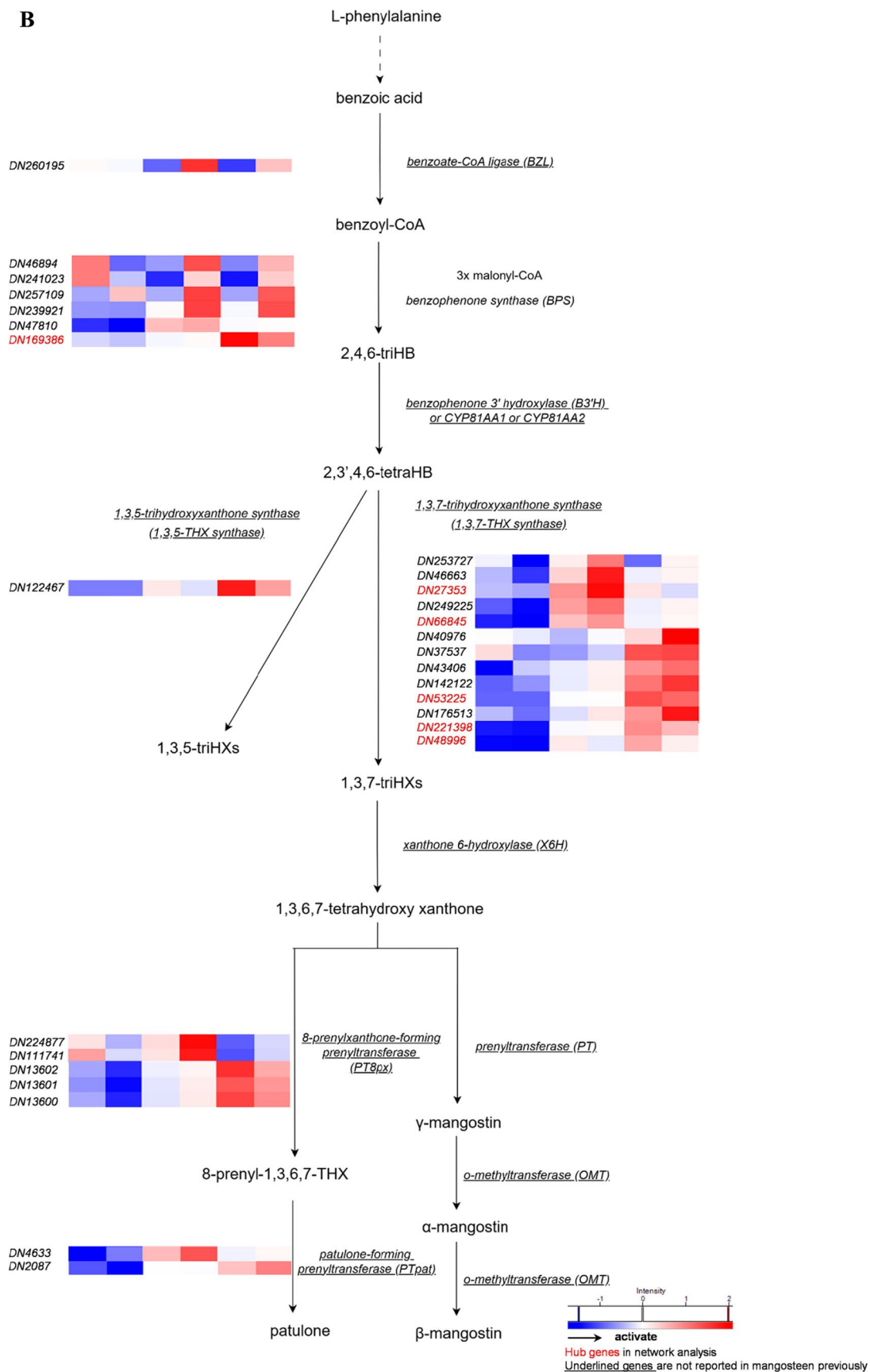
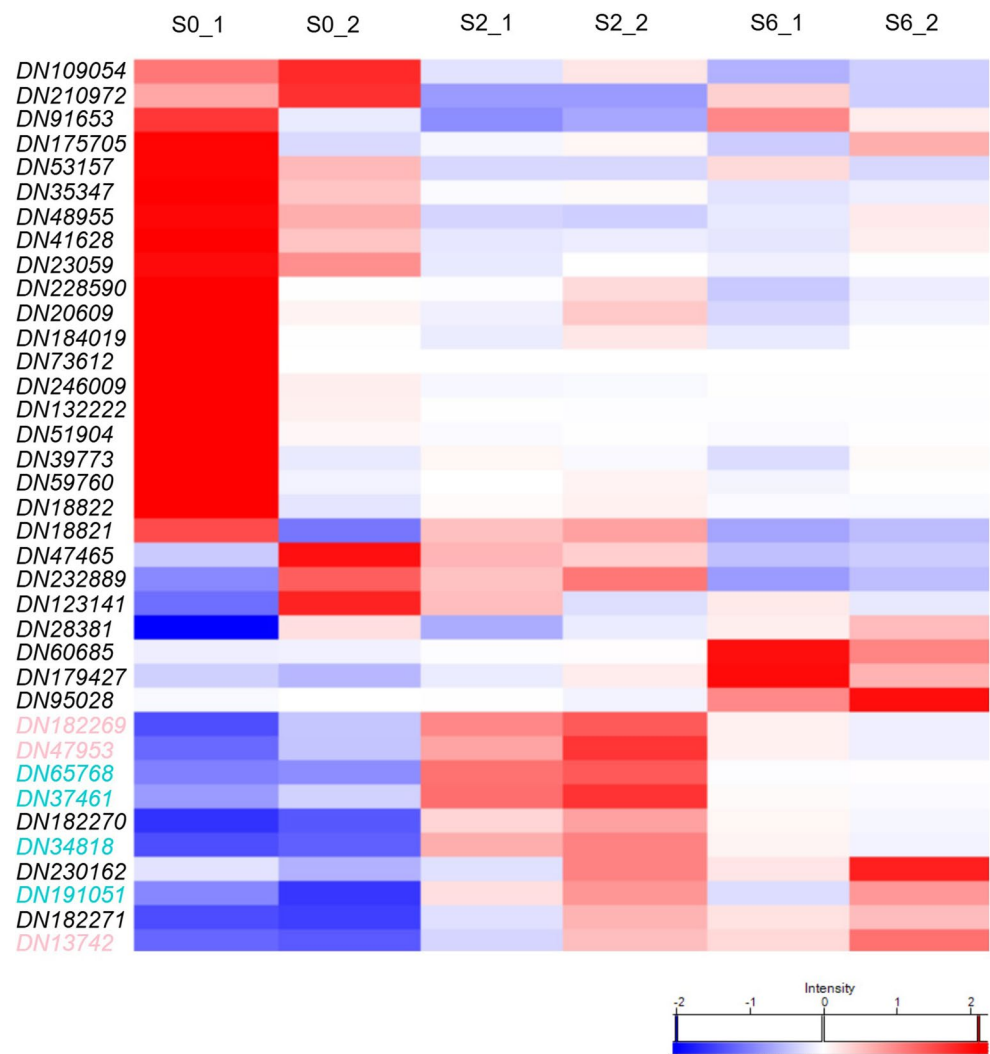


Fig. 4 (continued)

Fig. 5 Heatmap illustrates the expression levels of the 37 R2R3-MYBs genes identified from the network analysis across three mangosteen fruit ripening stages (S0, S2, and S6), based on Z-score normalized transcript per million (TPM) values. Genes belonging to the darkturquoise and pink modules are indicated by their respective font colors



5) with DN34818 ($\log_2FC=3.42$, $FDR=2.06E-20$) and DN13742 ($\log_2FC=2.36$, $FDR=2.04E-06$) showing significant upregulation at this stage (Supplementary Files 3–4).

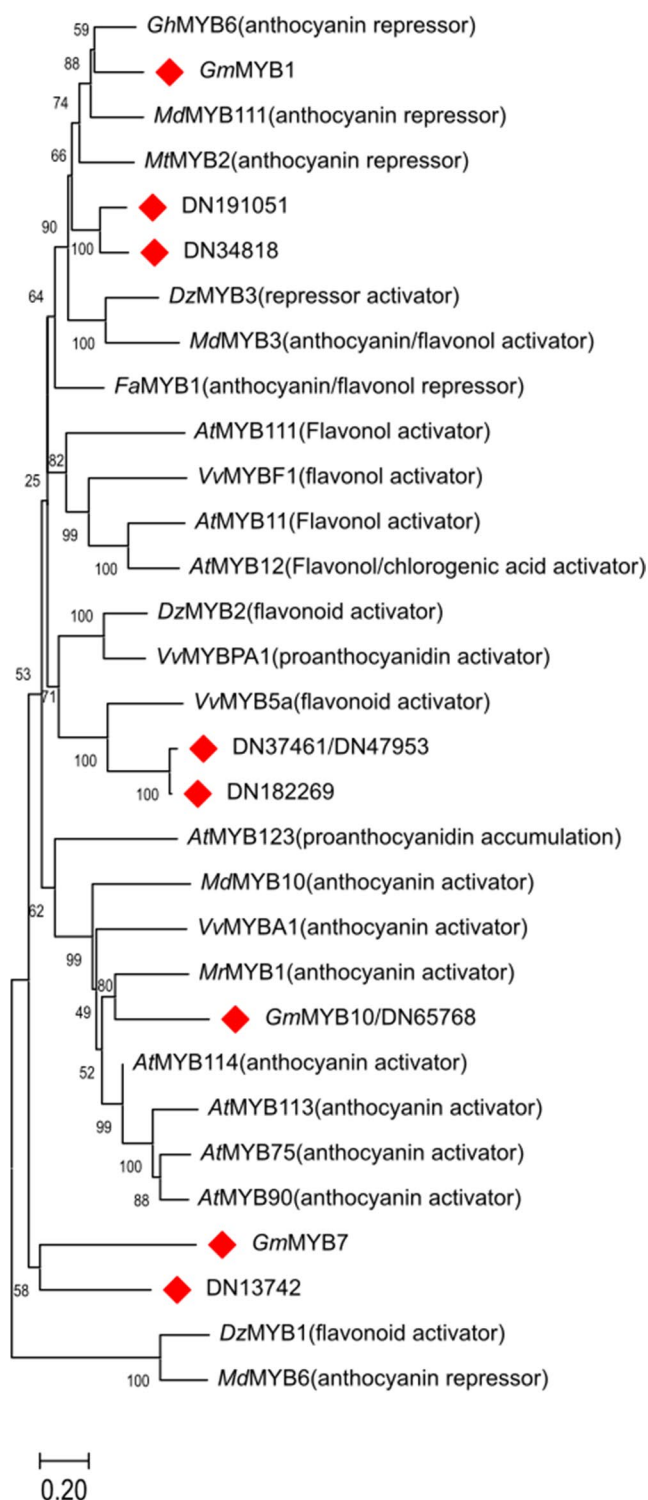
Phylogenetic Analysis of R2R3-MYB Proteins in Modules of Interest

A phylogenetic tree was constructed using the protein sequences of R2R3-MYBs identified from modules of interest in mangosteen and the key regulators (activators or repressors) involved in flavonoid biosynthesis (Deng et al. 2018; Liu et al. 2015; Pandey et al. 2015) using MEGA 11 (Fig. 6). *GmMYB1*, *GmMYB10* and *GmMYB7*, previously reported in an earlier study (Palapol et al. 2009a, b), were also included in the analysis. These MYBs were named according to the subgroup classification system described by Stracke et al. (2001).

Phylogenetic analysis from the previous study showed that *GmMYB1* is composed of 242 amino acids and shares

high sequence homology with strawberry *FaMYB1*, a well-known repressor of anthocyanin biosynthesis (Aharoni et al. 2001). In contrast, *GmMYB7* and *GmMYB10* both encode proteins comprising 284 amino acids. That study also reported that *GmMYB7* clustered closely with several *Arabidopsis thaliana* MYB proteins (*AtMYB108*, *AtMYB78*, *AtMYB112*, *AtMYB116*, and *AtMYB62*), which belong to subgroup7, generally associated with biotic/abiotic stress responses and development (Abe et al. 2003; Devaiah et al. 2009; Mengiste et al. 2003). Meanwhile, *GmMYB10* was found to be more closely related to the *Arabidopsis* MYBs in subgroup 10 (*AtMYB75*, *AtMYB90*, *AtMYB113*, and *AtMYB114*), which are established activators of anthocyanin biosynthesis (Palapol, Yossapol et al. 2009).

In the present study, *GmMYB1*, DN191051, and DN34818 cluster with anthocyanin repressors, whereas DN37461/DN47953 and DN182269 group with flavanol activators. *GmMYB10*/DN65768 cluster with anthocyanin activators, while *GmMYB7* and DN13742 formed



a separate cluster, distinct from both anthocyanin and flavanol regulators (Fig. 6). As previously *GmMYB7* was reported associated with subgroup 7 MYBs, both *GmMYB7* and DN13742 might be specialized in biotic/abiotic stress responses and developmental regulation, rather than in flavonoid biosynthesis.

Fig. 6 Phylogenetic tree of R2R3-MYBs identified in modules of interest in *G. mangostana* with various plants, including *Arabidopsis* (*AtMYB11*; Q9LZK4, *AtMYB12*; Q22264, *AtMYB111*; Q9FJ07, *AtMYB123*; Q9FJA2, *AtMYB75*; Q9FE25, *AtMYB90*; Q9ZTC3, *AtMYB113*; Q9FNV9, *AtMYB114*; Q9FNV8), durian (*DzMYB1*; XP_022740009, *DzMYB2*; XP_022757570, *DzMYB3*; XP_022763061), grapevine (*VvMYB5a*; AAS68190, *VvMYBA1*; BAD18977, *VvMYBF1*; ACV81697, *VvMYBPA1*; CAJ90831), apple (*MdMYB3*; AEX08668, *MdMYB6*; AAZ20429, *MdMYB10*; ACQ45201, *MdMYB111*; ADL36754), strawberry (*FaMYB44*; XP_004287994, *FvMYB44*; XP_004305954, *FaMYB1*; AAK84064), Chinese bayberry (*MrMYB1*; ADG21957), Mexican cotton (*GhMYB6*; AAN28286), and Barrel medic (*MtMYB2*; XP_003616388). The figure was generated using MEGA 11 software with the Neighbor-Joining method and a bootstrap of 1,000 replicates. The amino acid sequences of mangosteen MYBs and the previously identified *GmMYBs* (*GmMYB1*, *GmMYB7*, and *GmMYB10*) are marked with the red shape

Co-expression Analysis of the R2R3-MYB TFs with Associated Transcriptional Regulators of Anthocyanin and Xanthone Biosynthesis

A hub gene is defined as a gene that is highly connected with other genes (Langfelder and Horvath 2008). In this study, hub genes were identified as DEGs with at least 90% of the highest degree of connectivity in each module of interest. Two MYB hub genes were identified: *DN34818* in the dark-turquoise module and *DN13742* in the pink module. Phenolic compound biosynthesis genes (including xanthone and anthocyanin) and transcription factors co-expressed with these hub genes are shown in Fig. 7. In addition to MYB TFs, several other key fruit ripening-related transcription factors were also detected as hub genes in the networks, including MADS (AP3), C2C2-CO-like TF COL9, YAB5, and ethylene-responsive (ERF). Although bHLH and WRKY TFs were not identified as hub genes, they were present in the network. Their presence is expected, as they typically participate in the MYB–bHLH–WD40 (MBW) complex, which is well known for its role in regulating anthocyanin biosynthesis.

Discussion

Fruit ripening is a complex process that requires the molecular coordination of genes and transcription factors. In mangosteen, the fruit rind undergoes a striking color transition from green to purple, dark purple, or black as it ripens (Fig. S1). This color transformation is accompanied by an increase in anthocyanin and total phenolic content, including xanthenes (Gondokesumo 2019; Jamil et al. 2021; Palapol et al. 2009a, b). Anthocyanins are the predominant pigments responsible for this change, synthesized through the shikimate, phenylpropanoid, and flavonoid pathways (Vogt 2010).

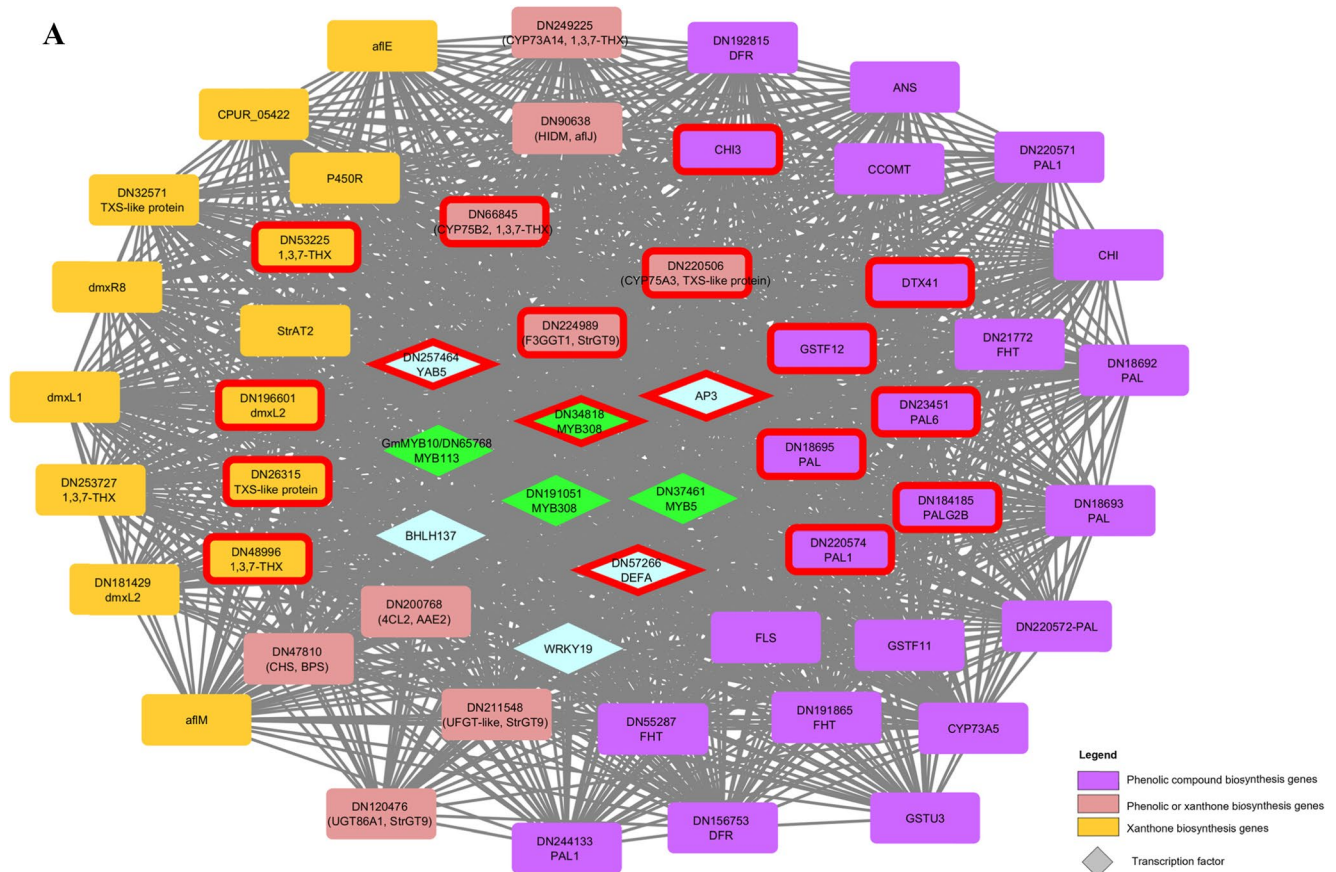


Fig. 7 Interaction subnetwork of phenolic and xanthone biosynthesis-related genes and transcription factors in (A) darkturquoise, (B) pink module, and (C) plum2 modules. R2R3-MYB transcription factors are highlighted as green diamond shapes and hub genes are indicated with red borders

MYB TFs are central regulators of anthocyanin biosynthesis across plants, including strawberries (Jiang et al. 2023), pears (Li et al. 2024), and berries (Lafferty et al. 2022). However, in mangosteen, only three MYB TFs, *GmMYB1*, *GmMYB7*, and *GmMYB10*, have previously been identified (Palapol et al. 2009a, b). Given the distinct presence of xanthenes in *Garcinia*, understanding how MYB TFs coordinate anthocyanin and xanthone biosynthesis remains a critical knowledge gap.

Using the recently published mangosteen transcriptome (Wee et al. 2024), a total of 91 R2R3 MYBs were identified in *G. mangostana* for the first time. Co-expression network analysis revealed three modules, darkturquoise, pink, and plum2, positively correlated with fruit ripening stages 2 and 6 (Fig. S5, S6). Seven MYBs were associated with darkturquoise and pink modules, both enriched for phenolic compound biosynthetic genes. Although no R2R3-MYBs were found in the xanthone-enriched plum2 module (Table 3), its close clustering with darkturquoise and pink modules (Fig. S6) suggests potential cross-module regulatory interactions. These relationships indicate that MYB TFs

may influence xanthone metabolism indirectly through correlated networks.

Phylogenetic analysis (Fig. 6) indicated that *GmMYB1*, DN191051, and DN34818 cluster with known anthocyanin repressors such as *FaMYB1* (Aharoni et al. 2001), *GhMYB6* (Wang et al. 2022), *MdMYB111* (Zhang et al. 2019a, b), *MtMYB2* (Jun et al. 2015). Both DN191051 and DN34818 carry the repressor motif LxLxL (Kagale and Rozwadowski 2011) at the C-terminal region (Fig. S7), providing further evidence for their potential repressor function. Interestingly, the expression of *DN34818* was significantly upregulated (Fig. 5, Supplementary File 3) during early fruit ripening, in parallel with anthocyanin accumulation, implying that it may participate in a feedback mechanism to prevent excessive anthocyanin accumulation. It is hypothesized that DN34818 could repress anthocyanin biosynthesis by competing with activator MYBs (e.g., *GmMYB10*/DN65768) for bHLH partners within the MYB-bHLH-WD40 (MBW) complex, thereby attenuating target gene activation once a threshold of anthocyanin accumulation is reached. Functional assays, such as CRISPR/Cas9 knockout of DN34818, would help

B

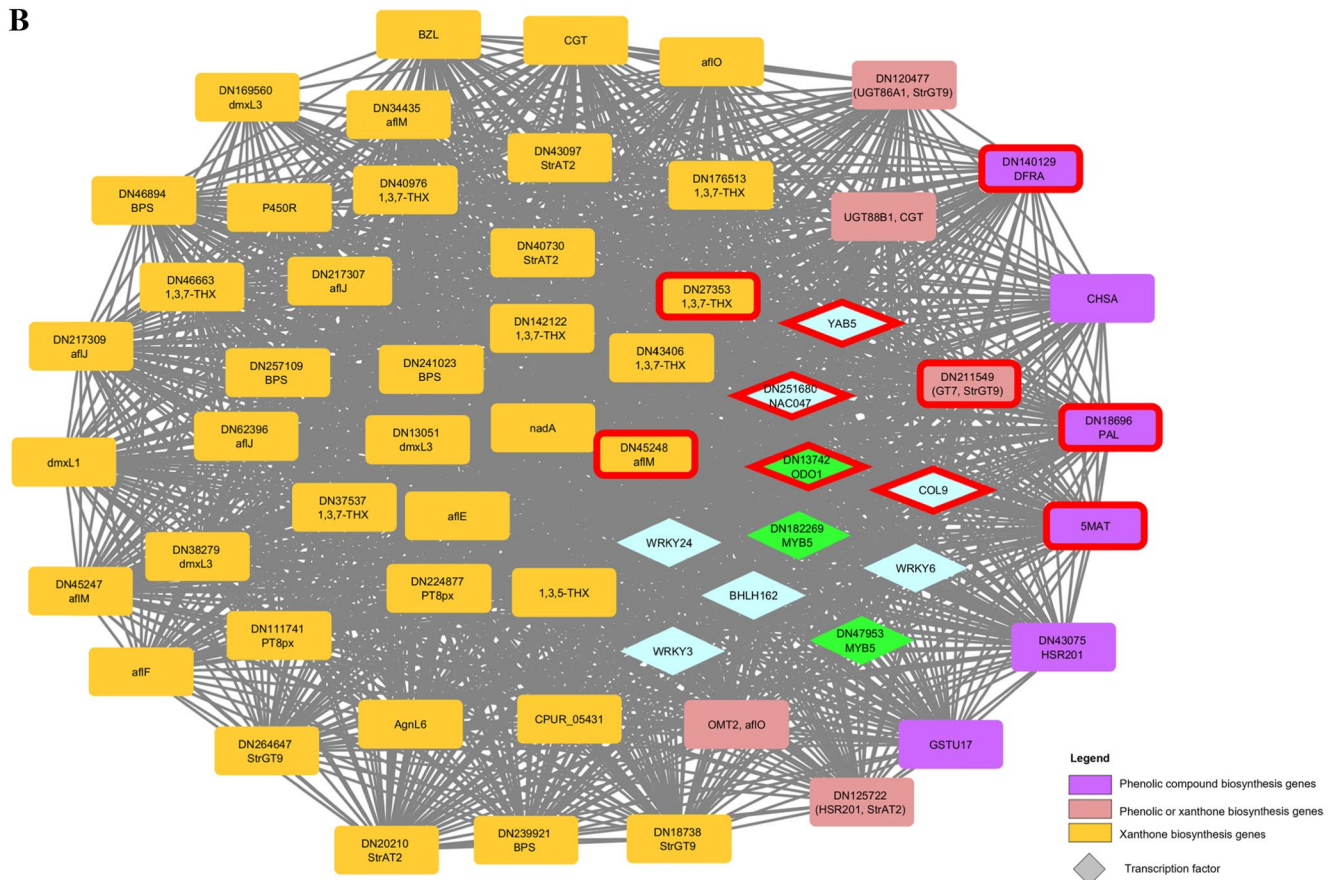


Fig. 7 (continued)

determine whether its loss results in anthocyanin hyperaccumulation and validate its role as a transcriptional repressor.

Conversely, *GmMYB10/DN65768*, previously shown in a transient dual luciferase assay to activate *GmDFR* and *AtDFR* promoters (Palapol et al. 2009a, b), clusters with anthocyanin activators such as *AtMYB75* (Zuluaga et al. 2008), *AtMYB90* (Borevitz et al. 2000; Gonzalez et al. 2008), *AtMYB113* (Gonzalez et al. 2008), and *AtMYB114* (Gonzalez et al. 2008). It showed increased expression at stage 2 of fruit ripening (Fig. 5), supporting its role as a positive regulator in mangosteen. Notably, all three genes (*DN65768*, *DN191051*, and *DN34818*) are part of the darkturquoise module, suggesting a coordinated regulatory balance in which activators and repressors act together to fine-tune anthocyanin levels during ripening, which was also reported in strawberry cultivar ‘Soelhyang’ (Hossain et al. 2018) and blueberry (Lafferty et al. 2022).

In addition, DN37461/DN47953 and DN182269 cluster with known flavanol activators such as *DzMYB2* (Weerawanich and Sirikantaramas 2024), *VvMYBPA1* (Bogs et al. 2007), and *VvMYB5a* (Deluc et al. 2008). These candidates may act as positive regulators of flavanol biosynthesis, similar to MYB5 homologs in strawberries (Jiang et al.

2023), pears (Li et al. 2024), and *Arabidopsis* (Dubos et al. 2010). Based on homology, these candidates could activate structural genes such as leucocyanidin reductase (LAR) and anthocyanidin reductase (ANR) that take part in flavanol accumulation. Although neither of them was detected in the current pink and dark turquoise modules, the presence of bHLH TFs in these modules suggests a possible MBW-like complexes that regulate flavanol and anthocyanin biosynthesis (Bulanov, Andrey N et al. 2025; Hichri et al. 2011).

A distinct clade formed by DN13742 and *GmMYB7* (Fig. 6) provides further insights into potential xanthone regulation. DN13742 is annotated as an ODORANT1 (ODO1) homolog, which in petunia controls phenylpropanoid metabolism and floral volatile production (Boersma et al. 2022). In mangosteen, DN13742 is in the pink module, closely related to the xanthone-enriched plum2 module (Fig. S6). Its increased expression during the early ripening stage suggests a role in initiating transcriptional reprogramming of phenylpropanoid pathways prior to xanthone accumulation. DN13742 may act as a regulatory bridge that enhances the supply of upstream phenylpropanoid intermediates toward xanthone biosynthesis. Functional assays such as over-expression or knockout analyses, coupled with promoter

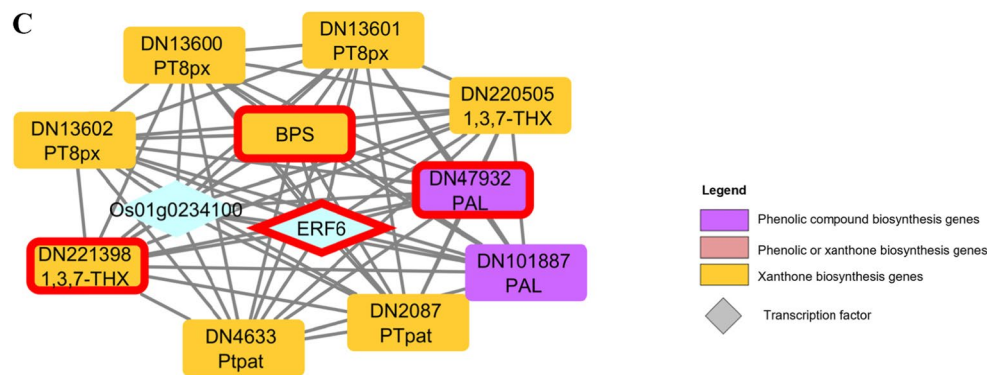


Fig. 7 (continued)

binding analyses of benzophenone synthase, would clarify its role in coordinating this metabolic transition.

Beyond MYBs, other transcription factors/regulators (FR-TF/TR) genes were identified as candidate hub genes across all three modules. These include *MADS* (*AP3*), *C2C2-CO-like TF* *COL9*, *YAB5*, and *ethylene-responsive (ERF)*. Functional evidence from other species supports their roles in phenolic biosynthesis. For instance, in *Artemisia annua*, *AaYABBY5* regulates the accumulation of anthocyanin and total flavonoid contents by targeting the promoters of *AaPAL*, *AaCHI*, *AaCHS*, and *AaUGT* (Kayani et al. 2021). Similarly, C2C2-CO-like TF was also involved in anthocyanin accumulation (Bai et al. 2014). *MADS*-MIKC TFs have a wide range of roles ranging from floral development (Li et al. 2022a, b; Saedler et al. 2001), fruit development and ripening (Vrebalov et al. 2002), fruit quality formation (Liu et al. 2018a, b), and root development (Alvarez-Buylla et al. 2019). ERF transcription factors are widely recognized for their regulatory roles during fruit ripening (Zhai et al. 2022). For instance, *FaRAV1*, an ERF gene in strawberries, binds to the promoter of *FaMYB10* to modulate anthocyanin biosynthesis (Zhang et al. 2020). Additionally, bHLH and WRKY transcription factors were present in both the darkturquoise and pink modules, which aligns with the established MYB-bHLH-WD40 (MBW) complex model that governs phenolic compounds biosynthesis (Bulanov et al., 2025a, b; Yan et al. 2021). The co-occurrence of these transcription factors supports the existence of coordinated regulatory networks that fine-tune the production of anthocyanins and xanthenes during mangosteen fruit ripening.

Finally, the integration of transcriptomics datasets from different tissues and developmental stages, generated using both Illumina and Ion Torrent platforms, increased the statistical power and enabled a broader meta-analysis. This integration allowed WGCNA to capture a wider range of expression variability, thereby improving the robustness and biological relevance of the identified co-expression modules. Despite the potential batch effects between

sequencing platforms, normalization of all samples based on TPM facilitated reliable cross-study comparisons, providing a comprehensive overview of transcriptional regulation during mangosteen ripening.

Overall, this study provides an initial framework for investigating MYB-mediated regulation of phenolic metabolism in mangosteen. The identification of candidate MYBs associated with anthocyanin and xanthone biosynthesis offers new molecular targets for future functional validation. This approach can also guide similar transcriptomic studies in other tropical fruits to uncover key transcriptional regulators of secondary metabolism.

Conclusion

This study identified 91 R2R3-MYB transcription factors in *G. mangostana* and revealed two novel candidates, DN34818 and DN13742, as potential key regulators within the anthocyanin and xanthone biosynthetic networks. Network and phylogenetic analyses suggest that DN34818 functions as a repressor that fine-tunes anthocyanin accumulation, while DN13742 may promote precursor channeling into xanthone production. These findings expand current knowledge of MYB-mediated regulation of phenolic metabolism in *G. mangostana* and provide a foundation for future genetic and biochemical validation.

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Author Contributions C.C.W., M.A. and H-H.G.: conceived and supervised the study. C.C.W. and I.A.R.: performed data analyses. C.C.W. and H-H.G.: wrote the manuscript. N.A.N.M., V.K.S. and M.A.: revised the manuscript. All the authors read and approved the final manuscript.

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Data Availability The Supplementary Files are available in Figshare (<http://doi.org/10.6084/m9.figshare.25038410>).

Declarations

Competing interests The authors declare no competing interests.

Generative AI The authors declared that the Generative AI ChatGPT-5 model was used to improve sentence clarity.

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